

Homework 3

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STAT651 - Statistical Theory I
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1 Question 1

To have your game end after n rolls you would need to have not gotten a 5 or a 6 $n - 1$ times in a row. For each of these $n - 1$ rolls the likelihood of not getting a 5 or a 6 is $4/6$. The likelihood of getting a 5 or 6 on a roll is $2/6$. Therefore, the likelihood of having to roll n times before getting a 5 or a 6 is:

$$P(x = n) = (4/6)^{n-1}(2/6)$$

This also means that:

$$P(x \leq n) = \sum_{i=0}^{n-1} (4/6)^i (2/6) = (2/6) \sum_{i=0}^{n-1} (4/6)^i$$

Therefore:

$$P(x > 3) = 1 - P(x \leq 3) = (2/6) = 1 - (2/6) \sum_{i=0}^{3-1} (4/6)^i \approx 0.2963$$

Therefore the probability that we need more than 3 tosses is 0.2963.

$$P(x \leq 2) = (2/6) \sum_{i=0}^{2-1} (4/6)^i \approx 0.5556$$

Therefore the probability that we need at most 2 tosses is 0.5556.

2 Question 2

We have $P(U|D) = 0.87$, $P(U|\neg D) = 0.73$, and $P(D) = 0.53$ where U is the belief that dog intelligence is underrated and D corresponds to have one or more dogs. We want to know $P(D|U)$. By Bayes theorem we have:

$$P(D|U) = \frac{P(U|D)P(D)}{P(U)}$$

The only thing we are missing here is $P(U)$. Given D and $\neg D$ form a partition:

$$\begin{aligned} P(U) &= P(U|D)P(D) + P(U|\neg D)P(\neg D) = P(U|D)P(D) + P(U|\neg D)(1 - P(D)) \\ &= 0.87(0.53) + 0.73(1 - 0.53) = 0.8042 \end{aligned}$$

Therefore:

$$P(D|U) = \frac{0.87(0.53)}{0.8042} \approx 0.5734$$

That is, if you believe dog intelligence is underrated there is a 57.34% probability that you own at least one dog.

3 Question 3

3.1 a

Premise: If $P(B) = 1$ then $P(A|B) = P(A)$ for any event A .

Proof: $P(A|B) = P(A \cap B)/P(B)$ which in this case is $P(A|B) = P(A \cap B)$ given $P(B) = 1$. Therefore what we'd like to prove is that $P(A \cap B) = P(A)$.

Assume $P(A \cap B^C) > 0$ then that would mean that $P(B^C) > 0$ as $P(B^C)$ must be greater than or equal to the probability of B^C intersected with some set. This in turn would mean that $P(S) = P(B) + P(B^C) > 1$ which cannot be. Therefore $P(A \cap B^C) = 0$. Given $P(A) = P(A \cap B) + P(A \cap B^C)$ we have $P(A) = P(A \cap B)$ and therefore that $P(A|B) = P(A \cap B)/P(B) = P(A|B) = P(A)/P(B) = P(A)$.

3.2 b

Premise: If $A \subset B$ then $P(B|A) = 1$ and $P(A|B) = P(A)/P(B)$.

Proof: If $A \subset B$ then $A \cap B = A$. Therefore $P(A \cap B) = P(A)$ which means that $P(A|B) = P(A \cap B)/P(B) = P(A)/P(B)$. Similarly $P(B|A) = P(A \cap B)/P(A) = P(A)/P(A) = 1$.

3.3 c

Premise: If A and B are mutually exclusive then:

$$P(A|A \cup B) = \frac{P(A)}{P(A) + P(B)}$$

Proof: A result we proved last homework is that:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

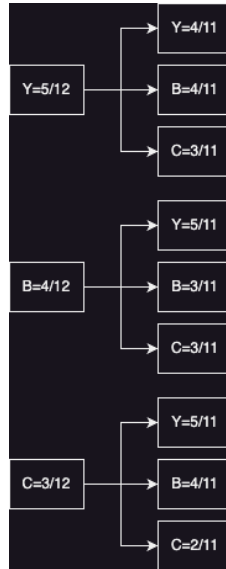
However in this case $P(A \cap B) = P(\emptyset) = 0$. Therefore $P(A \cup B) = P(A) + P(B)$.

It is also true that $A \cap (A \cup B) = A$, therefore $P(A \cap (A \cup B)) = P(A)$.

So it follows that:

$$P(A|A \cup B) = \frac{P(A \cap (A \cup B))}{P(A \cup B)} = \frac{P(A)}{P(A) + P(B)}$$

4 Question 4



- a $P(\text{Two Y's}) = (5/12)(4/11) = 20/132 \approx 0.1515$
- b $P(\text{Different Colors}) = 1 - P(\text{Same Colors}) = 1 - (5/12)(4/11) - (4/12)(3/11) - (3/12)(2/11) = 94/132 \approx 0.7121$
- c $P(\text{ the second puppy is Y } | \text{ the first puppy is C }) = 5/11 \approx 0.4545$
- d $P(\text{Same Colors}) = (5/12)(4/11) - (4/12)(3/11) - (3/12)(2/11) = 38/132 \approx 0.2879$
- e $P(\text{ the first puppy is C } | \text{ the second puppy is B })$. First let's get $P(\text{second puppy is B}) = (5/12)(4/11) + (4/12)(3/11) + (3/12)(4/11) = 44/132$. Next let's get $P(\text{ the first puppy is C and the second puppy is B }) = (3/12)(4/11) = 12/132$. Then $P(\text{ the first puppy is C } | \text{ the second puppy is B }) = (12/132)/(44/132) = 12/44 \approx 0.2727$
- f See above: $12/132 \approx 0.0909$

5 Question 5

First of all there are 50 choose 6 (15890700) possible hands. This will be our denominator in each of these cases.

5.1 a

1. Choose a denomination for your five of a kind - 10 options
2. Choose the singleton - 45 options (50 - the five of a kind)

Total = $(10)45 = 450$. Our probability is therefore:

$$\frac{450}{15890700} \approx 2.831(10^{-5})$$

5.2 b

1. Choose a denomination for your four of a kind - 10 options
2. Choose the four - 5 choose 4 = 5 options
3. Choose a denomination for your pair - 9 options
4. Choose the pair - 5 choose 2 = 10 options

Total = $(10)5(9)10 = 4500$. Our probability is therefore

$$\frac{4500}{15890700} \approx 2.831(10^{-4})$$

5.3 c

1. Choose a denomination for your four of a kind - 10 options
2. Choose the four - 5 choose 4 = 5 options
3. Choose the denominations of our singletons - 9 choose 2 = 36 options
4. Choose the card for the first singleton - 5 options
5. Choose the card for the second singleton - 5 options

Total = $(10)5(36)5(5) = 78750$. Our probability is therefore

$$\frac{78750}{15890700} \approx 4.956(10^{-3})$$

5.4 d

1. Choose a denomination for the pair - 10 options
2. Choose the pair - 5 choose 2 = 10 options
3. Choose the remaining denominations - 9 choose 4 = 126 options
4. For each singleton choose a card - $5^4 = 625$ options

Total = $(10)10(126)625 = 7875000$. Our probability is therefore

$$\frac{7875000}{15890700} \approx 0.4956$$

5.5 e

1. Choose your two denominations - 10 choose 2 = 45 options
2. Choose the cards for your first denomination - 5 choose 3 = 10 options
3. Choose the cards for your second denomination - 5 choose 3 = 10 options

Total = $45(10)10 = 4500$. Our probability is therefore

$$\frac{4500}{15890700} \approx 2.831(10^{-4})$$